Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/17/2006
PI Name:	Young, Gerald W Ph.D.		
Project Title:	Theory, Modeling, Software and Hardware Development in Computational Materials Science		
Division Name:	Physical Sciences		
Program/Discipline:	OTHER		
Program/Discipline--Element/Subdiscipline:	OTHER--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Akron	**State:**	OH
Zip Code:	44325-4002	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	06/07/2004	**End Date:**	03/03/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chait, Arnon	**Contact Phone:**	216-433-3558
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clemons, Curtis (University of Akron) HARIHARAN, S.I. (University of Akron)		
Grant/Contract No.:	NNC04GB27G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The focus of this Cooperative Agreement between the Computational Materials Laboratory (CML) of the Processing Science and Technology Branch of the NASA Glenn Research Center (GRC) and the Department of Theoretical and Applied Mathematics at The University of Akron is in the areas of system development of the CML workstation environment, modeling of microgravity and earth-based material processing systems, and joint activities in laboratory projects. These efforts complement each other as the majority of the modeling work involves numerical computations to support laboratory investigations.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	The original grant proposal was for the three year period, 6/7/04 - 6/6/07. The second year of the proposal was awarded but no funds were allocated by NASA due to NASA budgetary constraints. Nevertheless, by cutting back on personnel hours, the investigative team was able to provide modeling and system administrative support on site at NASA GRC as part of this cooperative agreement for a portion of the second year of the grant. This support continued until 10/31/05. At this point the grant funds were insufficient to continue support. Hence, personnel from the University of Akron no longer worked on site at NASA GRC after this date. Work continued at The University of Akron despite the lack of funding. This work focused on one-dimensional models of nanoscale oxidation. One-dimensional models are thought to be relevant in systems with a high relative humidity. In these systems a comparatively large surface layer of water covers the entire substrate. Our time-dependent, one-dimensional model accounts for an electric field (including the space charge build-up) coupled with the ion transport throughout the system. We have formulated and solved both high electric field and low electric field models where each model has three free boundaries. We have preliminary results for the electric field, ion concentration, and oxide thickness that qualitatively agree with the experimental observations. We demonstrate the influence of the space charge on the oxyde layer dynamics. In particular, we show both qualitatively and quantitatively how the growth of the oxide slows down with the increasing space charge. Our space charge formulation accounts for trapping of either positively or negatively charged ions. Further, we have included the possibility of generation of positively charged defects in the oxide. Such defects reduce the oxidation growth rate in two ways. First, the defects capture oxyanions and, second, the defects create an electrical field that acts in the direction opposite to that of the external field that drives the oxidation process. Our model also predicts the current in the nanocell system and directly correlates the normal component of the current density (integrated over time) with the oxide volume. This result is in excellent agreement with recently detected Faradaic current measurements during anodic oxidation.
Bibliography Type:	Description: (Last Updated: 04/03/2007)
Articles in Peer-reviewed Journals	Buldum A, Busuladzic I, Clemons CB, Dill LH, Kreider KL, Young GW, Evans EA, Zhang G, Hariharan SI, Keifer W. "Multiscale modeling, simulations, and experiments of coating growth on nanofibers. Part I. Sputtering." J Appl Physics. 2005 Aug 15;98:044303-10. , Aug-2005
Articles in Peer-reviewed Journals	Buldum A, Clemons CB, Dill LH, Kreider KL, Young GW, Zheng X, Evans EA, Zhang G, Hariharan SI. "Multiscale modeling, simulations, and experiments of coating growth on nanofibers. Part II. Deposition." J Appl Physics. 2005 Aug 15;98:44304-16. , Aug-2005
Articles in Peer-reviewed Journals	Marinov T, Buldum A, Clemons CB, Kreider KL, Young GW, Hariharan I. "Field emission from coated nanowires." J Appl Physics. 2005 Aug 15;98:44314-25 , Aug-2005